

Math 133 Written Homework 3

Due Thursday, September 24, 2026, at 11:59 p.m.

Submit one legible PDF on Gradescope, clearly identify each problem, and assign the correct page(s) to every problem. Show enough reasoning that another student in the course could follow your work, and include units whenever they are meaningful. You may discuss problems with classmates, but your submitted work must be your own. Late work will not be accepted. Use algebra, tables, and graphs; differentiation rules are not required. A calculator is allowed. Keep unrounded values in intermediate calculations and round monetary answers to the nearest cent unless instructed otherwise. The notation $\ln$ means the natural logarithm.

0. Assignment record.

- (a) List any collaborators and the problem(s) you discussed with each person. If none, write “None.”
- (b) How much time did you spend on this assignment?
- (c) Disclose any use of generative AI: name the tool and describe how you used it. If none, write “None.”

1. An investment model. The value of an investment is modeled by

$$A(t) = 2000(1.10)^t,$$

where $t \geq 0$ is time in years after the initial investment and $A(t)$ is measured in dollars. Assume no additional deposits or withdrawals.

- (a) Interpret 2000 and 1.10 in this model. Find the value of the investment after three years.
- (b) Use logarithms to solve $A(t) = 4000$. Give the doubling time to the nearest tenth of a year, and then give the first *whole-number year* when the model predicts a value of at least \$4,000.

2. **Total cost, average cost, and the next item.** The table gives a workshop's total cost $C(q)$, in dollars, for producing q items.

q (items)	0	10	11	12	20	30
$C(q)$ (dollars)	150	230	239	249	345	480

- Find the additional cost of the 11th item and of the 12th item. Write each answer as a difference of two function values and interpret it.
- Find the average rate of change of cost over $[0, 10]$, $[10, 20]$, and $[20, 30]$. Include units. Over which interval does cost increase fastest on average?
- Compute $C(10)/10$ and compare it with $C(11) - C(10)$. Explain the difference between *average cost per item* and the *additional cost of the next item*.

3. **From a difference quotient to a marginal estimate.** A producer models total cost, in dollars, by

$$C(q) = 100 + 4q + 0.10q^2, \quad 0 \leq q \leq 100.$$

Use the continuous model when q is not a whole number.

- Calculate $C(40)$ and $C(41)$. Find the exact additional cost of producing the 41st item.
- Calculate $[C(45) - C(40)]/5$. Interpret this number, including units. Explain why it need not equal your answer in part (a).
- Find the slope of the secant line between $(40, C(40))$ and $(40 + h, C(40 + h))$ where h is a variable. Write your answer as a fraction, and also simplify this when h is not equal to 0.
- Evaluate your simplified quotient for $h = 1$, $h = 0.1$, and $h = 0.01$, keeping three decimal places. Use the small-increment values to estimate the marginal cost rate at $q = 40$. Use that rate to estimate the cost of the 41st item, and compare with its exact cost from part (a).

4. **Interpreting marginal rates.** At a production level of 100 items, a business has total cost $C(100) = \$1,800$ and total revenue $R(100) = \$1,600$. Its marginal cost rate is \$8 per item and its marginal revenue rate is \$11 per item at this production level. These rates describe how the totals change for a small increase in production; they give *estimates* for additional whole items.
- Find $P(100) = R(100) - C(100)$ and the marginal profit rate. Interpret each, including units.
 - Estimate $C(101)$, $R(101)$, and $P(101)$. Assuming the marginal rates stay approximately constant over the next five items, estimate $P(105)$.
 - A manager says, “We are losing money at 100 items, so selling one more item must make the loss larger.” Do the marginal rates support this claim? Explain the distinction between the sign of profit and the sign of marginal profit.

5. **Does one more sale help?** A retailer models revenue and cost, in dollars, by

$$R(q) = 40q - 0.25q^2, \quad C(q) = 100 + 10q + 0.05q^2,$$

for $0 \leq q \leq 80$, where q is the number of items produced and sold. Profit is $P(q) = R(q) - C(q)$.

- Find a simplified formula for $P(q)$.
- For each starting quantity $q = 20$ and $q = 60$, calculate the exact one-item changes

$$R(q+1) - R(q), \quad C(q+1) - C(q), \quad P(q+1) - P(q).$$

Organize your results in a table. Verify that the profit change equals the revenue change minus the cost change.

- At each starting quantity, would producing and selling one more item increase or decrease profit? Explain using the additional revenue and additional cost, rather than the total revenue and total cost.
- Suppose fixed costs rise by \$100, so the new cost is $\tilde{C}(q) = C(q) + 100$. How do total profit and the one-item profit changes in part (b) change? Explain without recomputing the entire table.

6. **Excel investigation: marginal changes and a production decision.** A business models revenue and cost by

$$R(q) = \frac{2000q}{q+20}, \quad C(q) = 200 + 8q + 0.04q^2, \quad 0 \leq q \leq 100.$$

Here q is the number of items produced and sold, and revenue, cost, and profit $P(q) = R(q) - C(q)$ are measured in dollars. Use whole-number production levels.

In this problem, calculate *exact one-item changes*: the row for q should describe the change from q to $q+1$. These are the marginal changes for the *next* item.

- (a) **Build the table.** Put these headings in row 1:

Column	A	B	C	D	E	F	G
Heading	q	$R(q)$	$C(q)$	$P(q)$	ΔR	ΔC	ΔP

Fill A2:A102 with $0, 1, \dots, 100$. In row 2 enter

B2: =2000*A2/(A2+20)
C2: =200+8*A2+0.04*A2^2
D2: =B2-C2

Fill B2:D2 down through row 102. Then enter =B3-B2 in E2, =C3-C2 in F2, and =D3-D2 in G2. Fill E2:G2 down through *row 101 only*. Leave E102:G102 blank, because $q = 101$ is outside the model. Display monetary values to two decimal places.

- (b) **Interpret and check.** Report ΔR , ΔC , and ΔP at $q = 20$ and $q = 60$. What do these values say about producing the 21st item and the 61st item? Verify that E2-F2 agrees with G2, and check the same identity in a row with negative ΔP .
- (c) **Graph the changes.** Make two XY scatterplots with connected markers: one of $P(q)$ against q for $q = 0, \dots, 100$, and one showing both ΔR and ΔC against q for $q = 0, \dots, 99$. Label axes with units and identify the two series on the second graph.
- (d) **Choose a production level.** Use your table and graphs to find the whole-number quantity with the largest profit and report that profit. Check the one-item profit changes immediately before and after that quantity. Explain how the comparison of ΔR with ΔC accounts for the peak in the profit graph. Why is a positive marginal profit different from a positive total profit?

Include with Problem 6 in your Gradescope PDF: both labeled graphs; readable screenshots of rows for $q = 0, 1, 2$, $q = 20$, $q = 60$, and the rows around the profit maximum; the formulas entered in B2:G2; and your written answers.